H

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE NAME

CLASS

BIOLOGY

9648/03

Paper 3 Applications and Planning

September 2013

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use soft pencil for any diagrams, graphs or rough working.
Do no use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

Answer questions **1** to **4** in the spaces provided on the question paper.

Answer question **5** on the separate answer paper provided.

For Examiner's Use	
1	
2	
3	
5	
Total	60

For Examiner's Use	
4	12

This document consists of **15** printed pages and **1** blank page.

[Turn over

Section A

Answer **all** the questions in this section.

1 Bacteria can be genetically modified to produce insulin for human use. To achieve this, human insulin genes are transferred into bacteria. Plasmids containing two antibiotic resistance genes, one coding for resistance to tetracycline and one for resistance to ampicillin, are used to carry out this transfer.

A restriction enzyme was used to cut up the human DNA and plasmids. **Fig. 1.1** shows the different fragments of human DNA and the type of cut plasmid that was produced.

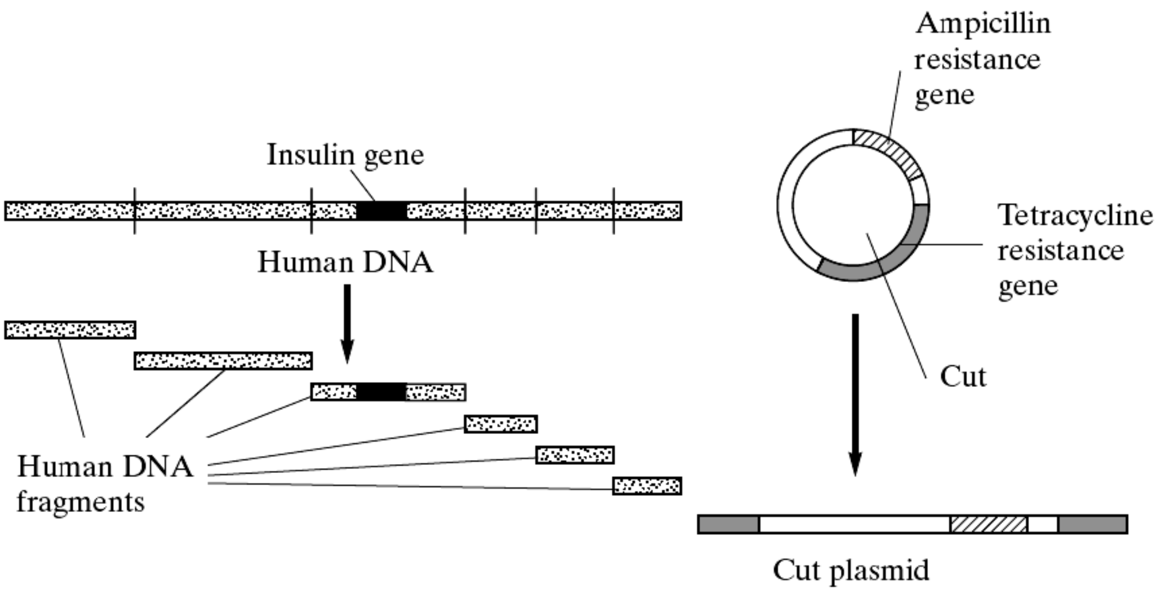

Fig. 1.1

(a) Describe a plasmid.

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....

....

.... [2]

(b) Suggest why the restriction enzyme has cut the human DNA in many places but has cut the plasmid DNA only once.

....

.... [2]

The fragments of human DNA and the cut plasmids were mixed together with DNA ligase. Several types of plasmid were formed. Some contained human DNA in the centre of the gene coding for resistance to tetracycline. The different types of plasmid are shown in **Fig 1.2**.

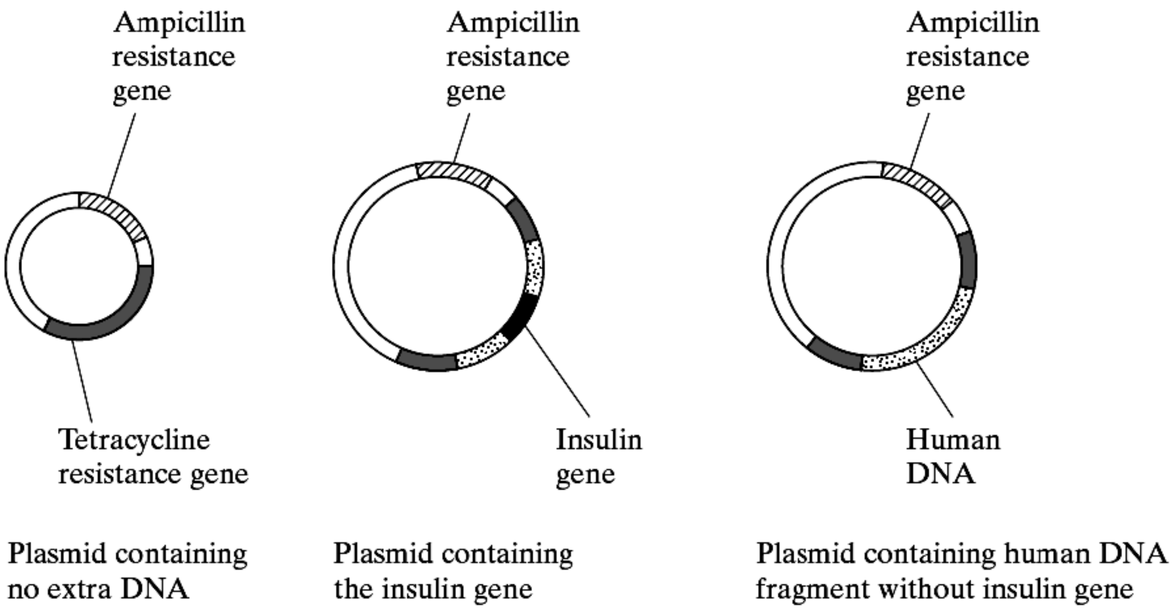

Fig 1.2

(c) Explain what causes several types of plasmid to be formed.

....

....

[2]

(d) Outline the reaction that DNA ligase catalyses.

....

[1]

The plasmids are mixed with the bacteria. Some bacteria take up the plasmids.

Some of the plasmids did not take up the extra human DNA. Replica plating was used to identify the bacteria with the human DNA. **Fig. 1.3** shows the bacterial colonies that grew on two replica plates.

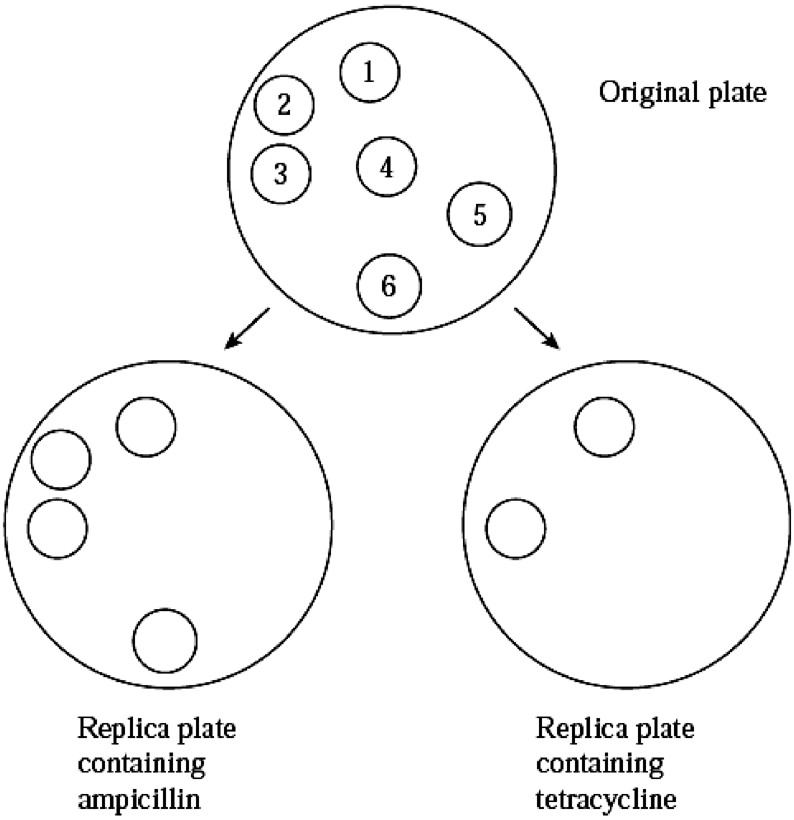

Fig. 1.3

(e) With reference to Fig. 1.3, explain the results of the replica plate containing:

(i) ampicillin.

....

....

....

.... [2]

(ii) tetracycline.

....

....

....

.... [3]

(f) Describe how the bacteria containing the insulin gene are used to obtain sufficient insulin for commercial use.

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....

....

[3]

[Total: 15]

2 Cord blood is the blood that circulates through the umbilical cord from the foetus to the placenta. The umbilical cord that remains attached to the placenta has been found to be rich in cord blood stem cells. Cord blood stem cells are the young or immature cells that can transform into other forms of essential blood cell types, such as red blood cells, white blood cells and platelets.

Adapted from: http://www.cordlife.com/sg/en/umbilical-cord-tissue/about-umbilical-cord-tissue

(a) Describe the unique features of all stem cells.

....

....

....
[3]

(b) Suggest an advantage of using cord blood stem cells in place of blood stem cells from the bone marrow.

....

....

....
[1]

(c) State the potency of cord blood stem cells.

....
[1]

There exist other groups of stem cells of greater developmental potential.

(d) Describe, with a named example, a group of stem cells with higher developmental potential than cord blood stem cells.

....

....

....
[2]

Besides using stem cell transplant to treat genetic disorders such as SCID, gene therapy could be another treatment option. Severe Combined Immunodeficiency (SCID) is a disease of young children in which patients have neither cell-mediated immune responses nor the ability to make antibodies.

There are two types of SCID: Adenosine Deaminase (ADA) deficient SCID and X-linked SCID. The delivery system used in SCID gene therapy is usually viral gene delivery system.

(e) Suggest a type of virus that can be used in SCID gene therapy.

....

[1]

(f) Explain the factors that keep gene therapy from becoming an effective treatment for SCID.

....

....

....

....

[3]

[Total: 11]

3 Golden rice is a transgenic variety of rice that has been genetically engineered to be rich in beta-carotene (a pre-cursor to vitamin A). **Fig. 3.1** below shows an artificial DNA sequence used in the production of golden rice.

Key:

pro – promoter sequence

ter – terminator sequence

Hyg resist – antibiotic resistance from a bacterium

Fig. 3.1

(a) Explain what is meant by *transgenic*.

....

[1]

(b) State the organism from which the promoter sequence for *Hyg* resist were taken from and explain your choice.

....

....

[2]

(c) Briefly outline one way in which this gene construct may be inserted into rice cells.

....

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....

[2]

(d) State why it is not possible to produce golden rice by breeding.

....

....

[1]

Hyg resist confers resistance to hygromycin, an antibiotic which kills both bacterial cells and plant cells.

(e) Explain the role of the *Hyg* resistance gene in this procedure.

....

....

[2]

(f) A concern of having *Hyg* resistance in the golden rice is the possible horizontal transfer of *Hyg* resist to other organisms.

(i) Suggest how *Hyg* resistance can be acquired by other plants and state a technical safeguard that can minimize this risk.

....

....

[2]

(ii) *"Hyg* resistance may result in a resistant strain of bacteria for which this antibiotic will no longer be effective." Explain if this claim is biologically valid.

....

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[2]

(g) Suggest 2 possible benefits of golden rice.

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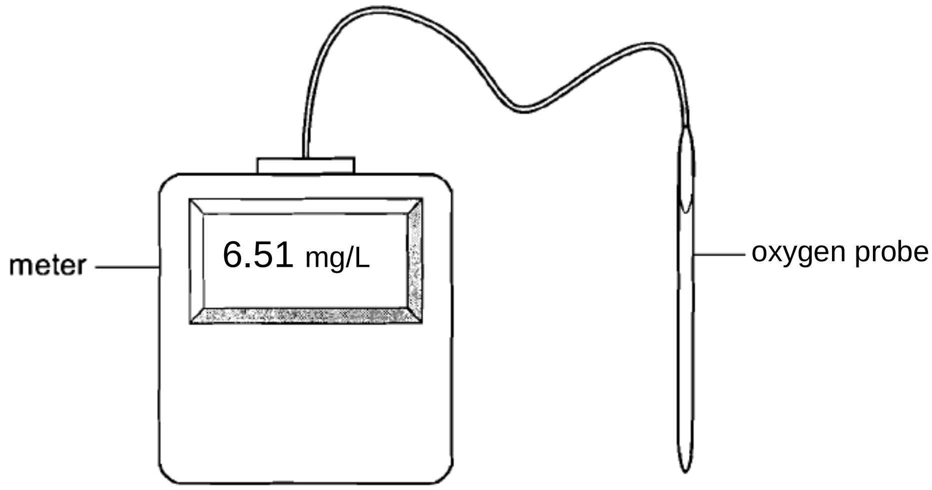
[2]

[Total: 14]

Planning Question

4 You are required to plan, but not carry out, an investigation into the effect of temperature on rate of photosynthesis.

Fig. 4.1 shows an oxygen probe and meter that can be used to measure the concentration of dissolved oxygen in a solution in mg/L.



Design an experiment, using an oxygen sensor to test the hypothesis that:

The rate of photosynthesis is dependent on the temperature.

Your planning can include, but is not limited to the following equipment & materials.

- A beaker of suspension of unicellular algae in water
- Bench lamp
- thermometer
- hot water at 80^{o}C
- stopwatch
- distilled water
- glass rod
- Stopwatch
- a variety of different sized beakers, test tubes, measuring cylinders, syringes, droppers and pipettes for measuring volumes

Your plan should include:

- an explanation of the theory to support your practical procedure,
- relevant, clearly labelled diagram(s) to show the arrangement of apparatus used,
- an explanation of dependent & independent variables involved,
- proposed layout of results tables and graphs with clear headings and labels,
- full details and explanations of the procedures you would adopt to ensure that the results obtained were as quantitative, precise and reliable as possible,
- correct use of scientific & technical terms

[Total: 12]

Free Response Question

Write your answers on the separate answer paper provided.
Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answers must be in continuous prose, where appropriate.
Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

5 **(a)** Using your knowledge of the genetic basis of cystic fibrosis, compare liposomal and viral delivery systems in the treatment of cystic fibrosis using gene therapy. [6]

(b) Describe how RFLP analysis can be used in genome mapping. [8]

(c) Discuss the ethical concerns that have arisen regarding the Human Genome Project. [6]

[Total: 20]